

# SUDONG LEE

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ORCID: [0000-0002-8928-6070](#)

## EDUCATION

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| <b>Ph.D. student in Robotics, Control, and Intelligent Systems (EDRS)</b><br><i>EPFL (École Polytechnique Fédérale de Lausanne)</i><br><i>Advisor: Prof. Josie Hughes</i>                                                                     | 2023.04. - present<br><i>Lausanne, Switzerland.</i> |
| <b>M.S. in Mechanical Engineering</b><br><i>Seoul National University</i><br><i>Advisor: Prof. Yong-Lae Park</i><br><i>Thesis: Modularized Robotic Skin Sensorized by Fiber Optic Force Sensing for Remote and Autonomous Robot Operation</i> | 2019.03. - 2021.08.<br><i>Seoul, Korea.</i>         |
| <b>B.S. in Mechanical Engineering</b><br><i>Korea University</i>                                                                                                                                                                              | 2013.03. - 2019.02.<br><i>Seoul, Korea.</i>         |

## RESEARCH EXPERIENCE

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| <b>CREATE Lab (Computational design of Robots and Embodied Autonomous TEchnologies Laboratory)</b><br>- <i>EPFL</i><br>Assistant-doctorant                                                                                                                                                                                                                                                                                                             | <i>Lausanne, Switzerland.</i><br>2023.04. - present                |
| <b>Soft Robotics Research Center (SRRC)</b><br>- <i>Seoul National University</i><br>Research Associate<br>Research Assistant<br>Research topics: <ul style="list-style-type: none"><li>· Fiber Jamming Actuator driven by Tendons to Enhance Adaptability</li><li>· Robotic Skin using 3-DoFs Force Sensor for Dexterous and Safe Interaction</li></ul>                                                                                               | <i>Seoul, Korea.</i><br>2022.09. - 2023.03.<br>2021.09. - 2022.08. |
| <b>Soft Robotics and Bionics Laboratory (SRBL)</b><br>- <i>Mechanical Engineering, Seoul National University</i><br>Graduate Student Researcher<br>Research topics: <ul style="list-style-type: none"><li>· Robotic Skin Sensorized by Fiber Optic Strain Sensors</li><li>· Multi-modal Locomotion and Environmental Adaptability of Legged Robots</li><li>· Soft Electronics and Sensors using Stretchable Materials and Sensing Mechanisms</li></ul> | <i>Seoul, Korea.</i><br>2019.01. - 2021.08.                        |

## HONORS AND AWARDS

|                                                                                                                                                                                                       |                                                                                                                                                                                                                                                         |
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| <b>M.S. Thesis Presentation Award</b><br>Mechanical Engineering, Seoul National University                                                                                                            | 2021.06.                                                                                                                                                                                                                                                |
| <b>Third Place Award for Locomotion Challenge</b><br>IEEE International Conference on Soft Robotics 2019 (RoboSoft 2019)<br>Team SRBL (Sudong Lee, G. Shin, J. Kim, M. Choi, Y. Baek, and Y.-L. Park) | 2019.04.                                                                                                                                                                                                                                                |
| <b>Great Honor, Winter 2018 Graduation</b><br>Korea University                                                                                                                                        | 2019. 02.                                                                                                                                                                                                                                               |
| <b>Semester High Honors</b><br>Korea University                                                                                                                                                       | 1 <sup>st</sup> Semester, 2018.,<br>2 <sup>nd</sup> Semester, 2017.,<br>1 <sup>st</sup> Semester, 2015.,<br>2 <sup>nd</sup> Semester, 2014.,<br>1 <sup>st</sup> Semester, 2014.,<br>2 <sup>nd</sup> Semester, 2013.,<br>1 <sup>st</sup> Semester, 2013. |

## SCHOLARSHIPS

**Kwanjeong Fellowship**  
Kwanjeong Educational Foundation

2<sup>nd</sup> Semester, 2020.,  
1<sup>st</sup> Semester, 2020.,  
2<sup>nd</sup> Semester, 2019.,  
1<sup>st</sup> Semester, 2019.

**National Science and Engineering Scholarship**  
Korea Student Aid Foundation

2<sup>nd</sup> Semester, 2018.,  
1<sup>st</sup> Semester, 2018.,  
2<sup>nd</sup> Semester, 2017.,  
1<sup>st</sup> Semester, 2015.

**Academic Excellence Scholarship**  
Korea University

2<sup>nd</sup> Semester, 2014.

**Best Honor Scholarship**  
Korea University

1<sup>st</sup> Semester, 2014.

## PUBLICATIONS

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### Journal Papers

1. **Sudong Lee**, K. Gilday, and J. Hughes, "Tendon-driven robotic shoulder joint with passive stabilization for large workspace," *IEEE/ASME Transactions on Mechatronics*, 2026. (DOI: [10.1109/TMECH.2026.3674717](https://doi.org/10.1109/TMECH.2026.3674717))
2. A. Georgopoulou, M. A. Calvo, L. Lucherini, **Sudong Lee**, J. Hughes, and E. Amstad, "Programmable somatosensory soft robots," *npj Flexible Electronics*, 2026. (DOI: [10.1038/s41528-026-00558-0](https://doi.org/10.1038/s41528-026-00558-0))
3. A. Georgopoulou, **Sudong Lee**, B. Dai, F. Bono, J. Hughes, and E. Amstad, "3D printing of self-healing longevous multi-sensory e-skin," *Communications Materials*, vol. 6, no. 121, 2025. (DOI: [10.1038/s43246-025-00839-7](https://doi.org/10.1038/s43246-025-00839-7))
4. **Sudong Lee**\*, J. I. Kim\*, Y. Baek, D. Chang, J. Lee, Y. S. Park, D. Lee, and Y.-L. Park, "Fiber-optic force sensing of modular robotic skin for remote and autonomous robot control," *IEEE Transactions on Robotics*, vol. 40, pp. 2373-2389, 2024. (DOI: [10.1109/TRO.2024.3378178](https://doi.org/10.1109/TRO.2024.3378178))  
\*: **Sudong Lee and J. I. Kim are co-first authors.**
5. D. Kim, **Sudong Lee**, T. H. Hong, and Y.-L. Park, "Exploration-based model learning with self-attention for risk-sensitive robot control," *npj Robotics*, vol. 1, no. 7, 2023. (DOI: [10.1038/s44182-023-00006-5](https://doi.org/10.1038/s44182-023-00006-5))
6. J. Kang\*, **Sudong Lee**\*, and Y.-L. Park, "Soft bending actuator with fiber-jamming variable stiffness and fiber-optic proprioception," *IEEE Robotics and Automation Letters*, vol. 8, no. 11, pp. 7344-7351, 2023. (DOI: [10.1109/LRA.2023.3316075](https://doi.org/10.1109/LRA.2023.3316075))  
\*: **Sudong Lee and J. Kang are co-first authors.**
7. T. Kim\*, **Sudong Lee**\*, S. Chang, S. Hwang, and Y.-L. Park, "Environmental adaptability of legged robots with cutaneous inflation and sensation," *Advanced Intelligent Systems*, 2300172, 2023. (DOI: [10.1002/aisy.202300172](https://doi.org/10.1002/aisy.202300172))  
\*: **Sudong Lee and T. Kim are co-first authors.**  
Cover Article: [10.1002/aisy.202370050](https://doi.org/10.1002/aisy.202370050), Editor's Choice: [\[Link\]](#)
8. Y. Lee, S. Lim, W. J. Song, **Sudong Lee**, S. J. Yoon, J.-M. Park, M.-G. Lee, Y.-L. Park, and J.-Y. Sun, "Triboresistive touch sensing: grid-free touch-point recognition based on monolayered ionic power generators," *Advanced Materials*, vol. 34, no. 19, 2108586, 2022. (DOI: [10.1002/adma.202108586](https://doi.org/10.1002/adma.202108586))
9. G. Shin\*, **Sudong Lee**\*, and Y.-L. Park, "Selective patterning of conductive elastomers embedded with silver powders and carbon nanotubes for stretchable electronics," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4983-4990, 2022. (DOI: [10.1109/LRA.2022.3153707](https://doi.org/10.1109/LRA.2022.3153707))  
\*: **Sudong Lee and G. Shin are co-first authors.**
10. T. Kim, **Sudong Lee**, T. Hong, G. Shin, T. Kim, and Y.-L. Park, "Heterogeneous sensing in a multifunctional soft sensor for human-robot interfaces," *Science Robotics*, Vol. 5, No. 49, eabc6878, 2020. (DOI: [10.1126/scirobotics.abc6878](https://doi.org/10.1126/scirobotics.abc6878))

### Conference Papers

1. **Sudong Lee** and J. Hughes, "Morphological and material programability of a hall-effect based soft tactile sensors," *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, San Diego, CA, USA, 2024, pp. 325-331. (DOI: [10.1109/RoboSoft60065.2024.10521990](https://doi.org/10.1109/RoboSoft60065.2024.10521990))

## CONFERENCE PRESENTATIONS

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1. A. Georgopoulou, **Sudong Lee**, B. Dai, J. Hughes, and E. Amstad, "Multimodal selective sensory receptors for robotic e-skin," *2025 IEEE 8th International Conference on Soft Robotics (RoboSoft)*, Lausanne, Switzerland, 2025.
2. **Sudong Lee\***, J. I. Kim\*, Y. Baek, D. Chang, J. Lee, Y. S. Park, D. Lee, and Y.-L. Park, "Fiber-optic force sensing of modular robotic skin for remote and autonomous robot control," *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Abu Dhabi, UAE, 2024.  
\*: **Sudong Lee and J. I. Kim are co-first authors.**
3. J. Kang\*, **Sudong Lee\***, and Y.-L. Park, "Soft bending actuator with fiber-jamming variable stiffness and fiber-optic proprioception," *2024 IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, 2024.  
\*: **Sudong Lee and J. Kang are co-first authors.**
4. G. Shin\*, **Sudong Lee\***, and Y.-L. Park, "Selective patterning of conductive elastomers embedded with silver powders and carbon nanotubes for stretchable electronics," *2022 IEEE 5th International Conference on Soft Robotics (RoboSoft)*, Edinburgh, United Kingdom, 2022.  
\*: **Sudong Lee and G. Shin are co-first authors.**

## PATENTS

1. T. Kim, **Sudong Lee**, and Y.-L. Park, "Soft Sensor with Multi-Sensing Function," 2022.  
(Korea Patent: 10-2384623)

## TEACHING EXPERIENCE

|                                                                                                                                                                   |                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Assistive Technology Challenge (ATC)</b><br>- HACKAHEALTH, EPFL<br>Teaching assistant, Instructor: Prof. Auke J. Ijspeert.                                     | Spring semester, 2024-25.       |
| <b>ME-320: Product development and engineering design</b><br>- Mechanical Engineering, EPFL<br>Teaching assistant, Instructor: Prof. Josie Hughes.                | Autumn semester, 2024-25.       |
| <b>Interdisciplinary robot competition</b><br>- EPFL<br>Team coach, Instructor: Prof. Auke J. Ijspeert.                                                           | Spring semester, 2023-24.       |
| <b>ME-320: Product development and engineering design</b><br>- Mechanical Engineering, EPFL<br>Teaching assistant, Instructor: Prof. Josie Hughes.                | Autumn semester, 2023-24.       |
| <b>M2794.001700_001: Mechanical product design</b><br>- Mechanical Engineering, Seoul National University<br>Teaching assistant, Instructor: Prof. Yong-Lae Park. | 1 <sup>st</sup> semester, 2019. |

## ACADEMIC SERVICE

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| Reviewer - Journal Papers<br><i>IEEE Transactions on Robotics (T-RO)</i><br><i>IEEE/ASME Transactions on Mechatronics (T-MECH)</i><br><i>IEEE Robotics and Automation Letters (RA-L)</i><br><i>Nature Communications</i><br><i>Soft Robotics</i>                |
| Reviewer - Conference Papers<br><i>IEEE International Conference on Robotics and Automation (ICRA)</i><br><i>IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i><br><i>IEEE International Conference on Soft Robotics (RoboSoft)</i> |

## TECHNICAL STRENGTHS (SKILLS)

|                                       |                                                                           |
|---------------------------------------|---------------------------------------------------------------------------|
| <b>Programming Languages</b>          | C++, Python, Matlab                                                       |
| <b>Embedded System</b>                | Arduino, AVR ATmega, Single-Board Computer (SBC)                          |
| <b>Software for System and Robots</b> | ROS, Pybullet                                                             |
| <b>Machine Learning</b>               | Pytorch, TensorFlow                                                       |
| <b>Design and Simulation</b>          | 3D Computer-Aided Design (CAD),<br>Finite Element Analysis (FEA) Software |
| <b>Fabrication</b>                    | 3D Printing (Additive Manufacturing), Silicone Fabrication                |

## OTHER EXPERIENCE

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**Republic of Korea Air Force (ROKAF, Military Service)**  
Staff Sergeant, Honorable discharge.

2015.08. - 2017.08.